

**23****Answer: A**

Since both branches have the same ratio vertically downwards, there is no current flowing through the middle horizontal wire.

$$\frac{1}{R_{eff}} = \frac{1}{10k + 30k} + \frac{1}{6.0k + 18k}$$

$$R_{eff} = 15 \text{ k}\Omega$$

**OR**

Since p.d. across the 10 k $\Omega$  and 6 k $\Omega$  resistors are the same, they can be treated as being a parallel connection.

Similarly the 30 k $\Omega$  and 18 k $\Omega$  resistors can be viewed as a parallel connection also.

$$R_{eff} = \left( \frac{1}{10} + \frac{1}{6.0} \right)^{-1} + \left( \frac{1}{30} + \frac{1}{18} \right)^{-1}$$
$$= 15 \text{ k}\Omega$$

The 1<sup>st</sup> method only works if resistors in both branches have the same ratio.  
The 2<sup>nd</sup> method always works.

**24****Answer: A**